

**LEARNSYNC: AN OFFLINE LAN-BASED LEARNING MANAGEMENT SYSTEM
WITH AI ACADEMIC ASSISTANT**

A Capstone Project

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I. INTRODUCTION

Education has always been a cornerstone of human progress, but in the 21st century, digital innovation has reshaped its delivery and accessibility. Cloud-based learning exemplifies this transition, enabling students and teachers to access resources anytime, anywhere, and on any device. Learners develop digital literacy skills such as online collaboration and resource management, while educators enhance their competencies through engaging the lessons. Jambhulkar (2025). Artificial intelligence has advanced education through personalized learning, adaptive feedback, and intelligent tutoring systems. However, these benefits often rely on cloud infrastructure, which disadvantages students in regions with poor internet access and deepens digital inequality. Offline AI models could help bridge this gap, but current solutions either sacrifice accuracy or require high computational power, making them unsuitable for low-resource environments. The key challenge is developing offline systems that balance efficiency and accuracy for practical, widespread use in education. Masum et.al (2025).

The research will be conducted at Palangon High School, located in Barangay Palangon Municipality of Libmanan, Province of Camarines Sur. The school is a public secondary institution under the supervision of the Department of Education (DepEd) Division of Camarines Sur with Mr. Roberto B. Malazarte HT-II/ School head. It caters to senior high school students from Palangon and neighboring barangays, serving as the primary educational institution the area. Palangon High School provides a range of academic programs aligned with the K-12 curriculum, Including core subjects and specialized strands for senior high school. For this study, the identified client Ma. Christina L. Quijano, A grade 12 teacher at Palangon High School, who plays a vital role in guiding

senior high school students and supporting the integration of innovative learning approaches such as e-learning.

Palangon High School currently facing problem about low internet connectivity and weak signal in the area. Limited access to stable online platforms further restricts the use of digital tools, hinders communication, and prevents the effective integration of technology into teaching and learning. Teachers are required to manually reproduce and distribute printed modules, a process that consumes substantial time and financial resources while straining the school's limited budget. The absence of a centralized system for managing digital files forces reliance on fragmented platforms such as USB drives, email, and messaging applications. This disorganized approach complicates access to essential instructional materials, increases the likelihood of duplication or misplacement, and delays the smooth execution of academic activities. The Palangon High school also lacks timely mechanisms to identify struggling students, resulting in delayed academic interventions and widening learning gaps.

The researcher wants to improve the learning gap by introducing LearnSync, an offline LAN-based Learning Management System equipped with an AI academic assistant that directly addresses the challenges faced by Palangon High School. Unlike cloud-dependent platforms, LearnSync operates through a local server within the school premises, allowing teachers and students to connect via LAN-enabled devices without relying on internet connectivity, thereby ensuring uninterrupted access to digital resources even in areas with weak or unstable signals. By centralizing instructional materials, teachers can upload lesson modules, quizzes, and activity sheets to the system, which students can then access smoothly. The AI academic assistant embedded in LearnSync

is designed to function as a lightweight, curriculum-aligned support tool trained on school-approved modules, enabling students to search and retrieve content efficiently, receive adaptive feedback personalized to their performance, engage in interactive questioning and guided explanations that simulate a tutoring experience, and benefit from generated summaries and highlighted key concepts that reduce cognitive overload and promote comprehension. At the same time, the assistant supports teachers in the early identification of struggling learners by analyzing student activity patterns and quiz results, allowing timely interventions to close learning gaps. By integrating these features, the academic assistant not only enhances student learning but also reduces teacher workload, making instructional delivery more efficient. LearnSync minimizes reliance on printed modules, saving both time and financial resources, while promoting sustainability and efficiency in teaching. Designed to be lightweight and scalable, the system can be implemented across classrooms and extended to community-based learning hubs, empowering Palangon High School to integrate digital innovation without the barriers of internet dependency. In this way, LearnSync provides a practical, cost-effective, and curriculum-aligned solution that enhances teaching and learning in low-resource environments. The AI academic assistant will be powered by the Mistral 7B model, ensuring efficient performance while maintaining accuracy in offline settings.

General Objectives

To design and develop cloud-based Learning Management System with an AI academic assistant that enables Senior High School students to access educational materials.

Specific Objectives

1. To develop an accessible learning management system that allows students to access modules, quizzes, and learning materials.
2. Enable teachers to upload modules, create quizzes, assign activities, and record student scores.
3. To Implement a local AI academic assistant that retrieves and generates responses based solely on teacher-uploaded materials.
4. To Provide a student performance monitoring module that enables teachers to track submissions, quiz results, and academic progress.

Scope and Delimitation

The scope of this project is the development of a LAN-based Offline AI Learning Assistant designed. It will allow teachers to upload text-based materials such as PDFs and Word documents, create quizzes, assign activities, and record student scores. Students will access these resources and interact with a local AI assistant that generates responses strictly from the uploaded content. A performance monitoring module will also be included to help teachers track submissions, quiz results, and academic progress, with access limited to mobile and desktop devices connected to the school's WiFi. For this study, the system will be implemented exclusively for Grade 12 students and will focus on the subject of Research, ensuring that the AI assistant provides curriculum-aligned support tailored to this specific academic requirement.

The delimitation of the project is that it will operate strictly offline, with no internet-based resources or external content. Access will be limited to the school's WiFi range,

excluding remote use. Supported materials will be restricted to text-based formats, excluding multimedia files such as videos. The AI assistant will only summarize and reframe uploaded content without generating new knowledge. The system will not include other grade levels such as Junior High and Grade 11.

Significance of the Study

The LearnSync system is expected to generate meaningful benefits to the following:

Teachers. The system helps by saving time and resources by uploading modules, creating quizzes, and monitoring student performance in one cloud-based platform.

Students. This study will have consistent access to the lesson and activities, with AI assistant that supports comprehension using only teacher upload materials.

Researcher. The study helps the researcher by enhancing technical expertise, contributing new knowledge to educational technology.

Future Researchers. The result of this study can use this project as a reference for integrating cloud-based systems and AI in education, guiding improvement and application.

Definition of terms

Learning Management System. The backbone of online education, streamlining the process of delivering and managing training programs. LMS serves as the central hub for Palangon High School's senior high students, streamlining distribution of modules, quizzes, and performance monitoring.

Local Area Network. Group of connected devices within a localized area. Allows students and teachers to access LearnSync through the school's WiFi without internet.

Offline System. software program that can function without a reliable and consistent internet connection. Runs on a local server at Palangon High School, allowing students and teachers to access materials, quizzes, and AI support without internet.

Artificial Intelligence Assistant. Helpful software application that uses artificial intelligence to perform tasks. It answers student questions based only on teacher-uploaded materials.

Performance Monitoring Module. Operations practice that automatically observes key metrics of application and system to ensure services are available and reliable. Helps teachers monitor quiz results, submissions, and student progress.

NOTES

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